

Supplementary Information

Taxonomic, functional and phylogenetic diversity of ant communities in response to elevational gradients in the Qinling Mountains

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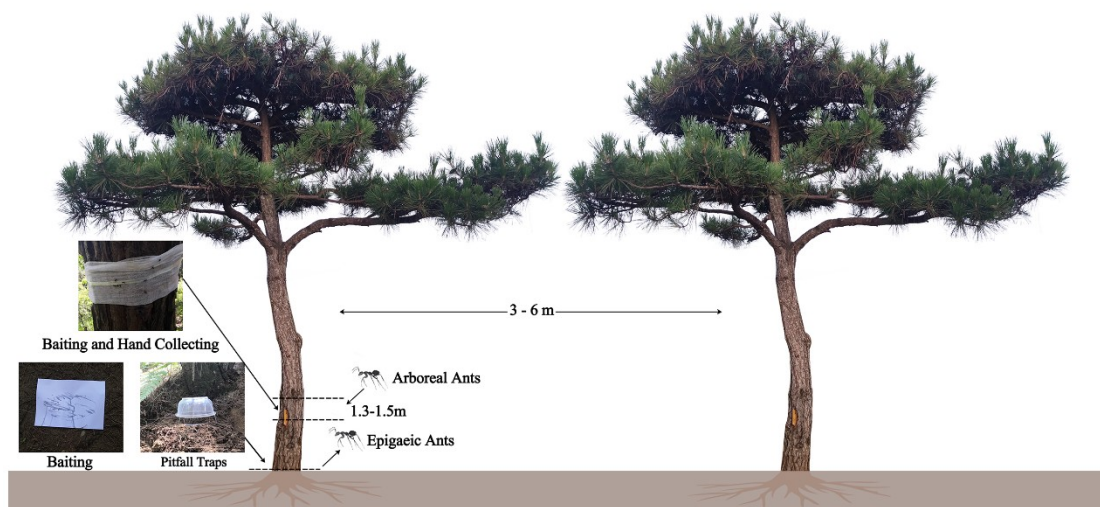


Fig. S1 The demonstration of sampling arboreal and epigaeic ants in this study.

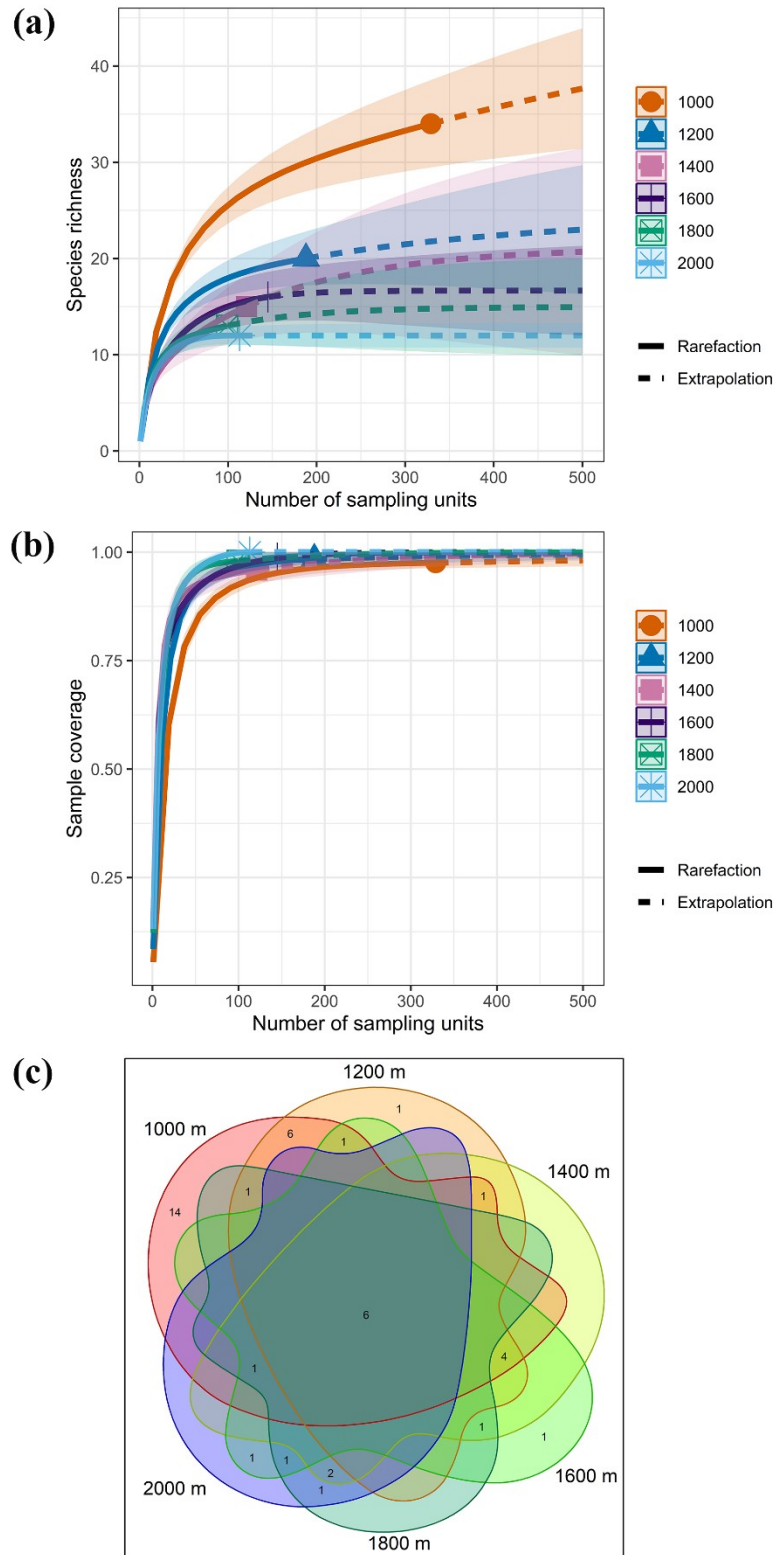


Fig. S2 (a) Species richness and **(b)** sample coverage of rarefaction and extrapolation (dotted line) curves with 95% confidence intervals (shaded areas) along different elevations; **(c)** Venn diagram illustrating the number of shared and unique ant species of each site along different elevations.

Table S1 Sampling design description

Elevational band	Site	Plot	Elevation	Main tree species
1000 m.a.s.l	ES1	EP1	967 m.a.s.l	<i>Pinus tabuliformis</i> , <i>Quercus</i> sp.
		EP2	1019 m.a.s.l	<i>Pinus massoniana</i> , <i>P. tabuliformis</i> , <i>Q.</i> sp.
		EP3	1046 m.a.s.l	<i>P. tabuliformis</i> , <i>Q.</i> sp.
1200 m.a.s.l	ES2	EP4	1198 m.a.s.l	<i>P. tabuliformis</i> , <i>Pinus armandii</i> , <i>Q.</i> sp.
		EP5	1256 m.a.s.l	<i>P. tabuliformis</i> , <i>Q.</i> sp.
		EP6	1298 m.a.s.l	<i>P. tabuliformis</i> , <i>P. armandii</i> , <i>Q.</i> sp.
1400 m.a.s.l	ES3	EP7	1448 m.a.s.l	<i>P. tabuliformis</i> , <i>P. armandii</i>
		EP8	1462 m.a.s.l	<i>P. tabuliformis</i> , <i>Q.</i> sp.
		EP9	1480 m.a.s.l	<i>P. tabuliformis</i>
1600 m.a.s.l	ES4	EP10	1616 m.a.s.l	<i>P. tabuliformis</i>
		EP11	1635 m.a.s.l	<i>P. tabuliformis</i> , <i>P. armandii</i>
		EP12	1643 m.a.s.l	<i>P. tabuliformis</i> , <i>P. armandii</i>
1800 m.a.s.l	ES5	EP13	1754 m.a.s.l	<i>P. tabuliformis</i>
		EP14	1803 m.a.s.l	<i>P. tabuliformis</i>
		EP15	1843 m.a.s.l	<i>P. tabuliformis</i>
2000 m.a.s.l	ES6	EP16	1915 m.a.s.l	<i>P. tabuliformis</i> , <i>Betula albosinensis</i>
		EP17	1997 m.a.s.l	<i>P. tabuliformis</i> , <i>Abies</i> sp., <i>Picea</i> sp.
		EP18	2029 m.a.s.l	<i>P. armandii</i> , <i>A.</i> sp., <i>P.</i> sp.

Note: m.a.s.l represents meters above sea level.

Table S2 The information of collected samples: ant species, the numbers of workers at each site and the relative abundance of each species

Subfamily	Ant species	Relative abundance (%)	Worker number at each site					
			ES1 (1000 m)	ES2 (1200 m)	ES3 (1400 m)	ES4 (1600 m)	ES5 (1800 m)	ES6 (2000 m)
Dolichoderinae	<i>Dolichoderus sibiricus</i> (Emery, 1889)	0.42	45	3	0	0	0	0
	<i>Liometopum sinense</i> (Wheeler, 1921)	2.77	314	0	0	0	0	0
	<i>Tapinoma geei</i> (Wheeler, 1927)	0.84	63	32	0	0	0	0
	<i>Tapinoma indicum</i> (Forel, 1895)	0.43	33	16	0	0	0	0
Formicinae	<i>Camponotus herculeanus</i> (Linnaeus, 1758)	0.03	0	0	0	3	0	0
	<i>Camponotus japonicus</i> (Mayr, 1866)	0.04	5	0	0	0	0	0
	<i>Camponotus mitis</i> (Smith, 1858)	0.21	10	6	8	0	0	0
	<i>Camponotus</i> sp. (Mayr, 1861)	0.02	2	0	0	0	0	0
	<i>Formica fusca</i> (Linnaeus, 1758)	0.33	0	0	12	0	13	13
	<i>Formica japonica</i> (Motschoulsky, 1866)	8.84	900	84	0	0	19	0
	<i>Lasius alienus</i> (Foerster, 1850)	9.72	859	106	1	35	57	45
	<i>Lasius flavus</i> (Fabricius, 1782)	0.24	4	1	1	21	0	0
	<i>Lasius fuliginosus</i> (Latreille, 1798)	8.48	110	832	0	20	0	0
	<i>Nylanderia pieli</i> (Santschi, 1928)	0.47	16	30	3	4	0	0
	<i>Nylanderia sharpii</i> (Forel, 1899)	20.70	602	518	554	642	30	3
	<i>Nylanderia</i> sp. (Emery, 1906)	1.54	0	175	0	0	0	0
	<i>Polyrhachis lamellidens</i> (Smith, 1874)	2.49	283	0	0	0	0	0
	<i>Polyergus samurai</i> (Yano, 1911)	0.59	67	0	0	0	0	0
	<i>Prenolepis angularis</i> (Zhou, 2001)	2.57	0	0	20	271	1	0
Myrmicinae	<i>Aphaenogaster caeciliae</i> (Viehmeier, 1922)	7.63	166	444	75	146	23	12
	<i>Crematogaster anthracina</i> (Smith, 1857)	10.57	1198	1	0	0	0	0
	<i>Crematogaster osakensis</i> (Forel, 1900)	0.56	63	0	0	0	0	0
	<i>Lordomyrma bhutanensis</i> (Baroni Urbani, 1977)	0.94	83	0	3	13	4	4
	<i>Myrmecina pauca</i> (Huang et al., 2008)	0.02	2	0	0	0	0	0
	<i>Myrmica luteola</i> (Kupyanskaya, 1990)	0.12	0	0	0	1	0	13
	<i>Myrmica mifui</i> (Chen et al., 2016)	0.78	0	0	0	3	14	72
	<i>Myrmica</i> sp. (Latreille, 1804)	4.63	11	16	174	181	110	33
	<i>Myrmica taibaiensis</i> (Wei et al., 2001)	3.00	0	0	0	0	35	305
	<i>Perissomyrmex bidentatus</i> (Zhou & Huang, 2006)	0.04	0	0	1	0	1	3
	<i>Pheidole fervida</i> (Smith, 1874)	6.41	583	35	88	15	2	4

Ponerinae	<i>Pristomyrmex punctatus</i> (Smith, 1860)	1.12	127	0	0	0	0	0
	<i>Strumigenys lewisi</i> (Cameron, 1886)	0.04	4	0	0	0	0	0
	<i>Sylophopsis</i> sp. (Santschi, 1915)	0.02	1	1	0	0	0	0
	<i>Temnothorax argentipes</i> (Wheeler, 1928)	0.07	8	0	0	0	0	0
	<i>Temnothorax shannxiensis</i> (Zhou et al., 2010)	0.95	1	45	28	11	9	14
	<i>Tetramorium smithi</i> (Mayr, 1879)	0.60	68	0	0	0	0	0
	<i>Brachyponera chinensis</i> (Emery, 1895)	0.01	1	0	0	0	0	0
	<i>Ectomomyrmex javanus</i> (Mayr, 1867)	1.24	123	18	0	0	0	0
	<i>Hypoponera beppin</i> (Terayama, 1999)	0.36	1	23	16	1	0	0
	<i>Hypoponera sauteri</i> (Wheeler, 1929)	0.02	2	0	0	0	0	0
	<i>Odontomachus monticola</i> (Emery, 1892)	0.01	1	0	0	0	0	0
	<i>Ponera paedericera</i> (Zhou, 2001)	0.13	7	5	1	2	0	0

Note: The relative abundance of each species = (Worker number of each ant species / total worker number of all ant species) ×100%

Table S3 The six functional traits measured, their measurement and functional significance

Functional trait	Trait measurement	Functional significance
Weber's length	The maximum length from the anterior edge of the pronotum to the posterior edge of the propodeum	It is the indicator of whole-body size, which is associated with foraging preference and defense strategies (Weiser & Kaspari, 2006).
Head width	In full face view across the eyes	Size of interstices through which ants can pass (Sarty et al., 2006); mandibular musculature (Kaspari, 1993).
Scape length	Collar to antennal elbow	It limits the distance ahead of the ant that it receives chemosensory information, for example pheromone trails (Yates et al., 2014).
Eye length	The maximum length of the eye	As an indicator of habitat use and foraging efficiency, eye length is related to vision location and food searching (Keller, 2011; Gibb et al., 2015).
Mandible length	From mandibular insertion to most distal point	It is related to the type of food resources consumed by ants (Weiser & Kaspari, 2006).
Hind leg length	The sum of hind femur length and hind tibia length	Longer legged ant species are characterized by efficient locomotion and foraging (Gibb & Parr, 2013).

Table S4 The data of six functional traits of all ant species

Subfamily	Species	Functional traits (mm)					
		Weber's length	Head width	Scape length	Eye length	Mandible length	Hind leg length
Dolichoderinae	<i>D. sibiricus</i>	1.160 ± 0.054	0.801 ± 0.042	0.710 ± 0.048	0.202 ± 0.013	0.426 ± 0.034	1.438 ± 0.077
	<i>L. sinense</i>	1.821 ± 0.243	1.343 ± 0.177	1.073 ± 0.184	0.245 ± 0.027	0.679 ± 0.131	2.572 ± 0.300
	<i>T. geei</i>	0.791	0.610	0.682	0.183	0.390	1.255
	<i>T. indicum</i>	0.490 ± 0.013	0.370 ± 0.004	0.380 ± 0.014	0.096 ± 0.006	0.204 ± 0.004	0.704 ± 0.006
Formicinae	<i>C. herculeanus</i>	3.387	2.653	2.162	0.566	1.231	5.509
	<i>C. japonicus</i>	2.837 ± 0.129	1.455 ± 0.163	1.906 ± 0.094	0.412 ± 0.048	0.887 ± 0.068	4.399 ± 0.313
	<i>C. mitis</i>	2.955 ± 0.220	1.501 ± 0.189	1.938 ± 0.055	0.461 ± 0.023	0.857 ± 0.032	5.169 ± 0.960
	<i>C. sp.</i>	1.668	1.092	1.173	0.269	0.534	2.445
	<i>F. fusca</i>	1.978 ± 0.184	1.201 ± 0.121	1.404 ± 0.111	0.433 ± 0.038	0.583 ± 0.052	3.205 ± 0.332
	<i>F. japonica</i>	2.428 ± 0.218	1.328 ± 0.115	1.700 ± 0.105	0.452 ± 0.045	0.747 ± 0.052	4.141 ± 0.413
	<i>L. alienus</i>	1.235 ± 0.070	0.976 ± 0.061	0.960 ± 0.036	0.227 ± 0.012	0.499 ± 0.028	1.991 ± 0.135
	<i>L. flavus</i>	1.335 ± 0.038	1.037 ± 0.036	1.015 ± 0.047	0.158 ± 0.008	0.617 ± 0.013	2.217 ± 0.057
	<i>L. fuliginosus</i>	1.331 ± 0.064	1.161 ± 0.069	1.106 ± 0.053	0.219 ± 0.020	0.513 ± 0.049	2.686 ± 0.132
	<i>N. pieli</i>	0.718 ± 0.032	0.503 ± 0.015	0.662 ± 0.022	0.122 ± 0.011	0.311 ± 0.017	1.215 ± 0.046
	<i>N. sharpii</i>	0.752 ± 0.040	0.517 ± 0.036	0.636 ± 0.023	0.126 ± 0.018	0.311 ± 0.022	1.187 ± 0.051
	<i>N. sp.</i>	0.740 ± 0.039	0.538 ± 0.039	0.670 ± 0.035	0.145 ± 0.015	0.324 ± 0.014	1.250 ± 0.089
	<i>P. lamellidens</i>	2.461 ± 0.271	1.756 ± 0.117	2.266 ± 0.146	0.410 ± 0.039	0.888 ± 0.062	5.194 ± 0.276
	<i>P. samurai</i>	2.412 ± 0.086	1.461 ± 0.080	1.222 ± 0.032	0.359 ± 0.039	0.874 ± 0.026	3.842 ± 0.109
	<i>P. angularis</i>	0.824 ± 0.015	0.613 ± 0.024	0.763 ± 0.043	0.162 ± 0.008	0.387 ± 0.022	1.649 ± 0.060
Myrmicinae	<i>A. caeciliae</i>	1.533 ± 0.111	0.949 ± 0.064	1.105 ± 0.109	0.192 ± 0.022	0.575 ± 0.052	2.283 ± 0.120
	<i>C. anthracina</i>	0.957 ± 0.053	0.919 ± 0.051	0.663 ± 0.036	0.190 ± 0.014	0.442 ± 0.033	1.467 ± 0.088
	<i>C. osakensis</i>	0.658 ± 0.047	0.592 ± 0.030	0.419 ± 0.016	0.131 ± 0.009	0.320 ± 0.016	0.884 ± 0.071

Ponerinae	<i>L. bhutanensis</i>	0.803 ± 0.025	0.598 ± 0.031	0.460 ± 0.016	0.111 ± 0.005	0.359 ± 0.035	0.886 ± 0.043
	<i>M. pauca</i>	0.711 ± 0.042	0.617 ± 0.064	0.424 ± 0.022	0.077 ± 0.001	0.368 ± 0.034	0.813 ± 0.098
	<i>M. luteola</i>	0.769	1.343	1.014	0.245	0.750	2.282
	<i>M. mifui</i>	1.723 ± 0.140	1.260 ± 0.034	1.143 ± 0.052	0.231 ± 0.011	0.628 ± 0.056	2.443 ± 0.084
	<i>Myrmica</i> . sp.	1.417 ± 0.098	0.944 ± 0.049	0.945 ± 0.066	0.201 ± 0.016	0.542 ± 0.025	2.002 ± 0.096
	<i>M. taibaiensis</i>	1.820 ± 0.148	1.271 ± 0.059	1.129 ± 0.025	0.238 ± 0.020	0.698 ± 0.013	2.433 ± 0.108
	<i>P. bidentatus</i>	0.756	0.851	0.656	0.089	0.556	1.315
	<i>P. fervida</i>	0.766 ± 0.061	0.614 ± 0.010	0.574 ± 0.009	0.126 ± 0.008	0.382 ± 0.012	1.059 ± 0.036
	<i>P. punctatus</i>	0.723 ± 0.056	0.785 ± 0.012	0.710 ± 0.035	0.154 ± 0.005	0.439 ± 0.017	1.241 ± 0.048
	<i>S. lewisi</i>	0.596	0.440	0.246	0.059	0.117	0.660
	<i>Sylophopsis</i> sp.	0.669	0.525	0.531	0.100	0.338	0.948
	<i>T. argentipes</i>	0.802	0.579	0.440	0.131	0.308	0.931
	<i>T. shannxiensis</i>	0.715 ± 0.057	0.562 ± 0.040	0.451 ± 0.008	0.136 ± 0.007	0.318 ± 0.011	0.863 ± 0.078
	<i>T. smithi</i>	0.961 ± 0.052	0.790 ± 0.015	0.544 ± 0.009	0.168 ± 0.007	0.449 ± 0.020	1.192 ± 0.020
	<i>B. chinensis</i>	1.287	0.754	0.759	0.141	0.480	1.581
	<i>E. javanus</i>	2.951 ± 0.148	1.908 ± 0.080	1.519 ± 0.122	0.281 ± 0.023	1.271 ± 0.033	3.317 ± 0.142
	<i>H. beppin</i>	1.129 ± 0.069	0.610 ± 0.031	0.579 ± 0.019	0.044 ± 0.007	0.418 ± 0.031	1.259 ± 0.038
	<i>H. sauteri</i>	0.579	0.341	0.310	0.013	0.261	0.600
	<i>O. monticola</i>	3.692 ± 0.203	2.153 ± 0.146	2.676 ± 0.173	0.302 ± 0.050	1.675 ± 0.085	5.343 ± 0.254
	<i>P. paedericera</i>	1.000 ± 0.063	0.546 ± 0.013	0.456 ± 0.006	0.032 ± 0.007	0.397 ± 0.008	0.894 ± 0.055

Note: The functional trait value of each species = Mean ± SD

Table S5 The percentage of subfamilies and genera in each site

Subfamily	Genus	The percentage of subfamily and genera in each site (%)					
		ES1 (1000 m)	ES2 (1200 m)	ES3 (1400 m)	ES4 (1600 m)	ES5 (1800 m)	ES6 (2000 m)
Dolichoderinae		7.90	2.13	0.00	0.00	0.00	0.00
	<i>Dolichoderus</i>	0.78	0.13	0.00	0.00	0.00	0.00
	<i>Liometopum</i>	5.45	0.00	0.00	0.00	0.00	0.00
	<i>Tapinoma</i>	1.67	2.01	0.00	0.00	0.00	0.00
Formicinae		49.59	73.27	60.81	72.75	37.74	11.71
	<i>Camponotus</i>	0.29	0.25	0.81	0.22	0.00	0.00
	<i>Formica</i>	15.62	3.51	1.22	0.00	10.06	2.50
	<i>Lasius</i>	16.88	39.27	0.20	5.55	17.92	8.64
	<i>Nylanderia</i>	10.72	30.24	56.55	47.19	9.43	0.58
	<i>Polyrhachis</i>	4.91	0.00	0.00	0.00	0.00	0.00
	<i>Polyergus</i>	1.16	0.00	0.00	0.00	0.00	0.00
	<i>Prenolepis</i>	0.00	0.00	2.03	19.80	0.31	0.00
		40.17	22.67	37.46	27.03	62.26	88.29
Myrmicinae	<i>Aphaenogaster</i>	2.88	18.57	7.61	10.66	7.23	2.30
	<i>Crematogaster</i>	21.88	0.04	0.00	0.00	0.00	0.00
	<i>Lordomyrma</i>	1.44	0.00	0.30	0.95	1.26	0.77
	<i>Myrmecina</i>	0.03	0.00	0.00	0.00	0.00	0.00
	<i>Myrmica</i>	0.19	0.67	17.66	13.51	50.00	81.19
	<i>Perissomyrmex</i>	0.00	0.00	0.10	0.00	0.31	0.58
	<i>Pheidole</i>	10.12	1.46	8.93	1.10	0.63	0.77
	<i>Pristomyrmex</i>	2.20	0.00	0.00	0.00	0.00	0.00
	<i>Strumigenys</i>	0.07	0.00	0.00	0.00	0.00	0.00
	<i>Sylophopsis</i>	0.02	0.04	0.00	0.00	0.00	0.00
	<i>Temnothorax</i>	0.16	1.88	2.84	0.80	2.83	2.69
	<i>Tetramorium</i>	1.18	0.00	0.00	0.00	0.00	0.00
		2.34	1.92	1.73	0.22	0.00	0.00
	<i>Brachyponera</i>	0.02	0.00	0.00	0.00	0.00	0.00
Ponerinae	<i>Ectomomyrmex</i>	2.13	0.75	0.00	0.00	0.00	0.00
	<i>Hypoponera</i>	0.05	0.96	1.62	0.07	0.00	0.00
	<i>Odontomachus</i>	0.02	0.00	0.00	0.00	0.00	0.00
		0.02	0.00	0.00	0.00	0.00	0.00

<i>Ponera</i>	0.12	0.21	0.10	0.15	0.00	0.00
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Note: The percentage of subfamily or genera in each site (%) = (Worker number of each subfamily or genera in each site / total worker number in each site) ×100%

Table S6 Summary of one-way ANOVA tests or Kruskal-Wallis test for ant diversity and functional traits composition among low, middle and high elevation. Shown were the mean $\pm$ sd and the p value.

	Low elevation	Middle elevation	High elevation	P value
Species richness	20.000 $\pm$ 4.517	10.833 $\pm$ 2.041	10.667 $\pm$ 0.516	0.0026 **
Chao-1	21.710 $\pm$ 4.382	13.955 $\pm$ 4.909	13.147 $\pm$ 3.024	0.0051 **
Rao' Q	2.524 $\pm$ 1.440	0.962 $\pm$ 0.388	0.955 $\pm$ 0.446	0.0148 *
FD_PD	12.730 $\pm$ 2.100	7.526 $\pm$ 1.331	8.151 $\pm$ 1.047	0.0001 ***
PD	1460.561 $\pm$ 239.020	859.856 $\pm$ 131.810	773.549 $\pm$ 37.987	0.0025 **
MPD	234.550 $\pm$ 7.675	217.287 $\pm$ 14.605	202.100 $\pm$ 4.190	0.0068 **
Weber's length CWM	1.292 $\pm$ 0.171	0.990 $\pm$ 0.068	1.533 $\pm$ 0.130	< 0.0001 ***
Head width CWM	0.938 $\pm$ 0.099	0.675 $\pm$ 0.042	1.076 $\pm$ 0.116	< 0.0001 ***
Scape length CWM	0.966 $\pm$ 0.124	0.763 $\pm$ 0.042	1.030 $\pm$ 0.062	0.0002 ***
Eye length CWM	0.212 $\pm$ 0.037	0.155 $\pm$ 0.009	0.225 $\pm$ 0.016	0.0023 **
Mandible length CWM	0.502 $\pm$ 0.052	0.400 $\pm$ 0.022	0.589 $\pm$ 0.056	< 0.0001 ***
Hind leg length CWM	2.127 $\pm$ 0.332	1.536 $\pm$ 0.110	2.198 $\pm$ 0.152	0.0002 ***

Asterisks (*) represents significant. Significant level: *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$.

Table S7 Summary of Wilcoxon signed-rank test for the statistical significance of the observed functional and phylogenetic diversity and the expected metrics of the null model in low, middle and high elevation. Shown were the p value.

	Low elevation	Middle elevation	High elevation
Rao' Q SES	0.5625	0.03125 *	0.03125 *
FD_PD SES	0.3125	0.3125	0.4375
PD SES	0.2188	0.4375	0.03125 *
MPD SES	0.1563	0.8438	0.03125 *

Asterisks (*) represents significant. Significant level: *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$.

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